Assignment – 1.16

Network Testing Tools Theory

Q - 1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

- **Ping www.youtube.com**
- **Command:**

  ping www.youtube.com

- **result:**
  - Packets sent: 4
  - Packets received: 4
  - Packets lost: 0
  - Average response time: 25 ms
- **Conclusion:** YouTube is reachable, and the average response time is 25 ms in this sample.

Q - 2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Hint: Each hop represents a network device your data passes through to reach the destination.

- **Traceroute www.flipkart.com**
- **Command on Windows:**

  tracert www.flipkart.com

- **output:**

Hop	Destination
1	192.168.1.1
2	10.20.0.1
3	172.16.0.1
4	203.0.113.10
5	Destination reached

- **Conclusion:** Traceroute shows the network hops between your computer and Flipkart. Actual hops may differ, and some may time out.

Q - 3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

- **Find the IP address of www.zomato.com**
- **Command:**

  nslookup www.zomato.com

- **result:**

  Name: www.zomato.com

  Address: IP address returned by DNS

- **Conclusion:** The nslookup command finds the IP address associated with the website name. Record the actual IP address shown on your computer.

Q - 4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

- **Ping a Local Device**
- **Command:**

  ping 192.168.1.5

- **Result:** If the device is online, it replies. If offline, it may show “Request timed out.”

<u>Q - 5.</u> Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.
Constraint: Give a specific example for each tool based on your own test results above.

- **<u>Short Report:</u>**

1. **Ping** checks whether a website or device is reachable.
2. Example: Ping YouTube to check its response time.
3. **Traceroute** shows the path taken by data to reach a website.
4. Example: Tracert Flipkart to see network hops.
5. **Nslookup** finds a website's IP address.
6. Example: Nslookup Zomato to check DNS information.